

are how the investments in private equity, real estate, infrastructure, and other nonfactor assets are funded. The alpha (net of costs) that CPPIB generates justifies having more than 900 employees in active management around the world in Toronto, London, and Hong Kong, rather than a bare-bones staff of fifteen managing a few index funds.

Panel C of Figure 14.5 is CPPIB's asset mix and exposure mix at March 31, 2013. At this date, CPPIB managed CAN\$183 billion. The fund holds 0.0% in foreign sovereign bonds, one of the Reference Portfolio's factors, but its economic exposure to foreign government bonds is 6.1%, which it achieves through its real estate, private equity, and infrastructure deals. Similarly, CPPIB holds 16.9% of its portfolio in real estate and infrastructure and yet it reports 0.0% economic exposure in these asset classes. For CPPIB, real estate and infrastructure are merely vehicles for obtaining Reference Portfolio factor exposure and are themselves not asset classes. Finally, CPPIB holds 34.9% of its portfolio in foreign developed market equities, but its economic exposure is 48.7%. The economic factor exposure in equities is far higher as it includes the implied equity exposure from real estate and infrastructure investments.

CPPIB's governance structure is crucial in allowing the organization to pursue factor investing and hire talented people (see Part III of the book for delegated investing). The Reference Portfolio means that management is "fully accountable for every dollar that we spend. . . . This keeps management focused and accountable," says Denison. CPPIB can do this partly because it is independent of political meddling. There are no ex-officio government officials on its board. CPPIB does not submit its investment strategy or business plans for government approval, nor does it need approval for its compensation policies. The independence structure of CPPIB is enshrined in the fund's enacting legislation and changing its independence charter is, in the words of Donald Raymond, its senior vice-president and chief investment strategist, "more difficult than changing the Canadian constitution."

When I've taught my case study on CPPIB, some quip that CPPIB's factor investing procedure is "complicated." Isn't it just easier to find and invest in a private equity deal? If you do that, you are implicitly giving yourself long equity exposure combined with short credit exposure. That is, you're getting the factor exposures anyway. The difference is that CPPIB handles and tries to explicitly control the factor risk. A major advantage of factor investing is that you know what the fundamental drivers are behind your assets and understand the circumstances under which your portfolio may do badly. It more accurately identifies alpha. During the financial crisis, many "alternative assets" tanked right along with equities; we should recognize these factor exposures *ex ante* rather than being surprised *ex post*. As the Professors' Report recommends, it is better to get a top-down handle on factor exposure than having it result haphazardly from ad hoc, bottom-up decisions.